

Benchmarking latent representations for future cell-state prediction in single-cell RNA-seq with neural SDEs

Tom Dauvé

Abstract

Over the past decade, many latent representations have been introduced to provide meaningful representations of single-cell RNA sequencing (scRNA-seq) data. As the field has evolved, new datasets have also been developed to address increasingly diverse tasks. One such task is the prediction of future cell states, including future gene expression counts. Several methods have been proposed to model cell evolution, including flow matching and neural differential equations. However, relatively little attention has been paid to the choice of representation in which these predictions are performed and evaluated.

This raises a natural question: which representation should be used for future cell-state prediction? While this question is well studied in the context of VAE models and reconstruction tasks, it is less clear whether the same conclusions hold for future prediction tasks.

We address this question by evaluating four representations—PCA, SCVI, VELOVI, and FlatVI—on the same future cell-state prediction task using the scDiffEq paradigm, which builds upon the approach introduced by PRESCIENT. scDiffEq models cell dynamics using a stochastic differential equation parameterized by a neural network and optimized through an optimal transport objective.

We focus on the LARRY dataset, a lineage-tracing dataset that provides clonal relationships between cells observed at different time points. Within this experimental framework, we introduce several ways of comparing representations that account for both the latent-space displacement generated by scDiffEq and the decoding step associated with each representation.

We show that PCA and SCVI representations achieve better performance than VELOVI and FlatVI when evaluated in gene-expression space, whereas the differences between representations are more nuanced when evaluated in their respective latent spaces.

1 Introduction

1.1 Single-cell state representation

Single-cell RNA sequencing (scRNA-seq) is a technology that provides high-dimensional gene expression measurements at the level of individual cells. Concretely, within a given tissue, the gene expression of each cell can be measured by counting the number of RNA molecules associated with each gene, including spliced RNA and, in some datasets, unspliced RNA. Considering all genes

expressed in a given species, this results in a matrix with dimensions $n_{\text{cells}} \times n_{\text{genes}}$. In practice, the number of cells can range from thousands to millions or even billions, while the number of genes typically ranges from approximately ten to fifty thousand, depending on the species and experimental setting.

This count matrix is highly sparse. Although each cell contains essentially the same genome, only a specific subset of genes is actively transcribed in a given cell, according to its cellular state and function. Consequently, only a small fraction of the possible gene-cell measurements correspond to detected RNA molecules.

Working directly with such a high-dimensional and sparse matrix is challenging, which has motivated the development of methods for representing cellular states in lower-dimensional spaces. Traditional dimensionality-reduction methods include Principal Component Analysis (PCA), while more recent approaches based on variational autoencoders (VAEs) have provided probabilistic latent representations specifically designed for single-cell data. Among these methods, scVI [1] has become an important framework for learning latent representations of scRNA-seq data.

1.2 Cellular dynamics

Understanding cellular biology requires not only characterizing the current state of cells, but also understanding how cellular states evolve over time. Predicting gene expression at the level of individual cells is a particularly challenging task. Two major difficulties arise.

First, future cellular states appear to be inherently stochastic and influenced by the surrounding environment. Two cells with very similar gene-expression profiles at a given time can follow very different trajectories and reach substantially different states in the future, either because they experience different microenvironments or due to intrinsic stochasticity. This is particularly relevant when studying undifferentiated cells, which can potentially differentiate into different cell types.

Second, it is currently not technically possible to directly measure the future gene expression of a specific living cell, since standard scRNA-seq measurements are destructive. To address this limitation, specific experimental protocols have been developed to provide information about the relationship between cells observed at different time points. Such datasets can therefore provide a form of correspondence between cell states at an initial time point and cell states observed at later time points.

1.3 Existing methods

In order to address this challenge, several methods have been proposed to model cellular dynamics and predict future cell states. A major difficulty is that we generally do not have an exact one-to-one correspondence between the initial and future states of individual cells. Instead, for a population of cells observed at an initial time point, we typically observe a distribution of cell states at a later time point.

Flow matching methods aim to learn a velocity field that transports the distribution of initial cell states toward the distribution of future cell states. PRESCIENT [2] and scDiffEq [3], in contrast, model cellular dynamics using neural stochastic differential equations to generate future cell states from an initial population. These approaches are closely related to the optimal transport problem, which provides a framework for relating cell populations observed at different time points.

1.4 Does the representation matter?

All of these approaches learn cellular dynamics within a specific representation of the cell state. This raises the question of whether the choice of representation affects the quality of future-state prediction. A representation that provides good reconstruction of observed cells is not necessarily well suited for modelling future cellular dynamics. scDiffEq was originally developed using PCA representations, and the authors explicitly discuss the possibility that PCA may not be the optimal representation for future cell-state prediction.

1.5 What we do

We investigate this question by comparing four representations: PCA, SCVI, VELOVI, and FlatVI. We use the LARRY dataset and the scDiffEq framework to benchmark these representations under the same future cell-state prediction task. We systematically evaluate different latent and model configurations across multiple random seeds. Predictions are evaluated using optimal transport in both latent space and gene-expression space.

We find that PCA and SCVI achieve better performance than VELOVI and FlatVI when predictions are evaluated in gene-expression space, whereas differences between representations are more nuanced when evaluated in latent space.

2 Background

2.1 Latent representations for scRNA-seq

As previously discussed, latent representations have been developed to address the high-dimensional and sparse nature of scRNA-seq data. The main idea is to project the high-dimensional gene-expression space into a lower-dimensional latent space using an encoder function. Working in this reduced space makes it easier to manipulate and model cellular states. A decoder can

then be used to map latent representations back to gene-expression space. However, since the latent space has a lower dimensionality than the original gene-expression space, the decoder cannot be considered a mathematical inverse of the encoder. Instead, it aims to reconstruct the relevant information from the latent representation.

In this subsection, we introduce the four latent representations used in this study: PCA, SCVI, VELOVI, and FlatVI.

2.1.1 PCA

Principal Component Analysis (PCA) [4] is one of the most widely used methods for dimensionality reduction. It consists of finding a set of orthogonal projection axes that maximize the variance of the projected data. The main advantage of PCA in this context is its linear, deterministic, and interpretable behavior, which makes it a natural baseline for latent-space representations.

2.1.2 SCVI

Single-Cell Variational Inference (SCVI) [1] belongs to the family of Variational Autoencoders (VAEs). It uses a neural network composed of an encoder and a decoder. The encoder maps the observed gene-expression profile of a cell to a probability distribution in the latent space, from which a latent representation is sampled. The decoder then uses this latent representation to model the observed gene expression.

Unlike PCA, SCVI therefore provides a stochastic latent representation. The decoder models gene expression using a probabilistic distribution designed to account for the statistical properties of scRNA-seq count data, including overdispersion. In particular, SCVI uses a Negative Binomial-based generative model.

For the evaluation performed in this study, we use the normalized expression output of the decoder, referred to as *px_scale*. This representation corresponds to the relative expression profile predicted by the decoder, independently of the library size of the cell.

SCVI is one of the most widely used approaches for learning latent representations of scRNA-seq data. It therefore provides an important reference point for the benchmark performed in this study.

2.1.3 VELOVI

Velocity Variational Inference (VELOVI) [5] extends the SCVI framework to model RNA velocity. Unlike SCVI, which is primarily designed to model gene-expression counts, VELOVI explicitly models the dynamics of gene expression by taking advantage of both spliced and unspliced RNA measurements.

VELOVI uses a stochastic variational autoencoder to infer latent representations together with parameters describing the underlying gene-expression dynamics. In particular, the model is based on a kinetic description of the relationship between unspliced and spliced RNA.

In this study, VELOVI is included because its latent representation is learned from information related to gene-expression dynamics, and may therefore contain

information relevant to future cell states. For the gene-expression-space evaluation, we decode the latent representations into expression profiles and normalize them before computing the optimal transport distance, so that the output can be compared with the other representations.

2.1.4 FlatVI

Flat Variational Inference (FlatVI) [6] addresses the geometry of the latent space learned by variational autoencoders. In particular, the geometry induced by the decoder can distort distances between cells when moving between latent space and gene-expression space. FlatVI therefore introduces a regularization term designed to make the geometry of the latent space closer to a Euclidean geometry.

FlatVI is based on the SCVI framework but adds this geometric regularization during training. The motivation for including FlatVI [6] in our benchmark is to investigate whether modifying the geometry of the latent space can improve future cell-state prediction with scDiffEq.

2.2 Modeling cellular dynamics

We consider cellular states observed at several time points, denoted by Z_{t_0} , Z_{t_1} , and Z_{t_2} . Importantly, these observations correspond to populations of cells rather than complete individual trajectories. In general, the exact path followed by a given cell between two consecutive measurements is therefore unknown. To model the evolution of cellular states between observed time points, we introduce neural differential equations.

2.2.1 Neural differential equations

Let $z(t)$ denote the latent state of a cell at time t . A first approach is to assume that the temporal evolution of this state follows an ordinary differential equation of the form

$$\frac{dz}{dt} = f(z, t).$$

If the function f is known, the cellular state can be propagated through time by integrating this differential equation. In a Neural Ordinary Differential Equation (Neural ODE), the function f is parameterized by a neural network and learned from data.

However, a purely deterministic formulation associates a unique future trajectory with a given initial state. This assumption may be too restrictive for cellular dynamics, since cells with similar initial states may evolve toward different future states. In order to account for this variability, the deterministic model can be extended to a stochastic differential equation (SDE),

$$dz = f(z, t) dt + g(z, t) dW_t,$$

where W_t denotes a Wiener process. The first term corresponds to the deterministic component of the dynamics and is usually referred to as the *drift*. The second term introduces stochasticity into the trajectory and is referred to as the *diffusion*.

The function $g(z, t)$ controls the magnitude of this stochastic component and can also be parameterized by a neural network. As a consequence, repeated simulations from the same initial state can produce different future states depending on the sampled stochastic trajectory. Neural SDEs therefore provide a natural framework for modelling multiple possible cellular futures from a common initial state.

Once future cell states can be generated, a second problem arises: we need a quantitative way to compare a predicted population of cells with the population that is actually observed at a future time point.

2.2.2 Optimal transport

Within the neural SDE framework, an initial population at time t_0 can be propagated to obtain a predicted population at time t_1 , denoted by $z_i^{\text{pred}}(t_1)$. This predicted population must then be compared with the experimentally observed population at the same time point, denoted by $z_j^*(t_1)$.

A direct cell-to-cell comparison is generally not appropriate. First, there is no known one-to-one correspondence between predicted and observed cells. Second, the two populations may contain different numbers of cells. The problem is therefore better formulated as a comparison between two distributions rather than between individual observations.

Optimal transport [7] provides a natural framework for this comparison. Intuitively, each cell is assigned a certain amount of mass, and the objective is to transport the mass of one distribution toward the other while minimizing the total transport cost. The mass associated with one point can be distributed among several target points, provided that the mass constraints of both distributions are respected.

More formally, let P and Q denote two distributions, and let γ_{ij} denote the amount of mass transported from point x_i in P to point y_j in Q . Given a cost function $c(x_i, y_j)$, the optimal transport problem can be written as

$$\text{OT}(P, Q) = \min_{\gamma \in \Pi(P, Q)} \sum_{i, j} \gamma_{ij} c(x_i, y_j),$$

where $\Pi(P, Q)$ denotes the set of admissible transport plans between the two distributions.

The resulting optimal transport cost provides a quantitative measure of the discrepancy between the predicted and observed cellular populations. A lower value indicates that less transport is required to transform one distribution into the other.

2.2.3 PRESCIENT and scDiffEq

PRESCIENT [2] and scDiffEq [3] are frameworks designed to model cellular dynamics from single-cell data using continuous-time stochastic dynamics. Their general principle is to learn a dynamical model that propagates an initial population of cells toward future states observed experimentally.

In the scDiffEq framework used in this study, cellular states are represented in a latent space and evolved

through a neural stochastic differential equation. Starting from cells observed at an initial time point, the model generates a population of predicted future states. The predicted population can then be compared with the experimentally observed future population using a distribution-level objective based on optimal transport.

This formulation is particularly suitable for single-cell data because the task does not require an exact one-to-one correspondence between cells observed at different time points. Instead, the model aims to reproduce the evolution of the population distribution over time.

2.3 Motivation for benchmarking latent representations

The representation methods introduced previously were originally developed with different objectives. PCA provides a simple, deterministic, and linear representation that is widely used across many applications. SCVI provides a probabilistic latent representation specifically designed to model scRNA-seq gene expression. VELOVI extends this framework to explicitly model RNA dynamics using spliced and unspliced RNA measurements. FlatVI, in contrast, focuses on the geometry of the latent space and aims to reduce distortions induced by the decoder.

These differences raise an important question. A representation that performs well for dimensionality reduction, reconstruction, or RNA-velocity modelling is not necessarily the representation in which future cellular dynamics are easiest to learn.

If the same dynamical framework is applied to different latent representations, differences in prediction performance can provide information about how the choice of representation affects the modelling of cellular dynamics. This motivates the central question of this study:

Does a representation that is suitable for representing or reconstructing the current cellular state also provide a suitable space for learning future cellular dynamics?

To investigate this question, we systematically apply the scDiffEq framework to PCA, SCVI, VELOVI, and FlatVI representations. For each representation, future-state prediction is first evaluated directly in latent space. Predicted latent states are then decoded back into gene-expression space, where prediction performance is evaluated again.

This two-level evaluation allows us to distinguish between the ability of scDiffEq to model dynamics within a given latent geometry and the ability of the resulting trajectories to produce biologically meaningful future gene-expression profiles after decoding. The benchmark therefore aims to assess the combined influence of latent-space geometry, representation learning, and decoding on future cell-state prediction.

3 Materials and Methods

3.1 Experimental overview

We use the LARRY dataset, which contains cell populations observed at different time points. We first train four latent representations from this dataset. We then train a scDiffEq neural network independently on each representation. Predictions are evaluated in the corresponding latent spaces and are then decoded into gene-expression space. The main objective of the experimental design is to keep the future-state prediction framework as consistent as possible across representations, while allowing each representation to use its corresponding decoder.

3.2 Dataset

3.2.1 LARRY lineage-tracing dataset

The LARRY dataset [8] was selected because it provides lineage information connecting cells observed at different time points. This information provides an experimental framework for studying changes in cell-state distributions over time.

The dataset contains 49,302 cells, 5,864 independent clones, and 23,420 genes from the species *Mus musculus*. Cells are divided into three time points: t_2 , t_4 , and t_6 . Both spliced and unspliced RNA measurements are available for each cell. We mainly use spliced RNA in this study, except for the VELOVI model, which requires both spliced and unspliced RNA measurements.

3.2.2 Temporal structure and lineage information

For training scDiffEq, we consider cells at t_2 as the initial population, while cells at t_4 and t_6 are considered future populations. The network is trained to predict the cell-state distributions at both t_4 and t_6 .

We use 80% of the independent clones for training and the remaining 20% for validation and analysis. The split is performed at the clone level to prevent cells belonging to the same clone from being present in both the training and validation sets.

No such split is applied during the training of the four latent representations. Each representation is trained using all available cells. The train/validation split is applied only to the subsequent scDiffEq experiments.

3.3 Data preprocessing

The following preprocessing steps are applied to both spliced and unspliced RNA count matrices.

3.3.1 General preprocessing

Genes expressed in fewer than three cells are removed. We then select the same set of 2,000 highly variable genes (HVGs) for all representations.

Using the same gene set across representations ensures that differences in downstream performance are not caused by differences in the input gene set.

3.3.2 Normalization

For SCVI, FlatVI, and VELOVI, we provide the preprocessed count matrices directly to the models, as these models explicitly model count data. For VELOVI, both unspliced and spliced count matrices are provided.

For PCA, the data are normalized using counts per million (CPM) with a target sum of 10^4 . A $\log 1p$ transformation is then applied. Finally, the expression values are centered and scaled using the gene-wise standard deviation.

3.4 Latent representation learning

3.4.1 Training

Once the data have been preprocessed as described above, each representation is trained using latent dimensions of 10, 20, 50, 100, and 200.

The PCA representation is obtained using the Scanpy API. SCVI and VELOVI models are trained using scvi-tools version 1.4.2. FlatVI is implemented using a custom implementation, as no official API was available for this method at the time of the study.

For FlatVI, we follow the procedure described in the original method to compute the metric tensor using Jacobian-vector products (JVPs). The main difference from the SCVI framework is the addition of the geometric regularization term. We set the corresponding regularization coefficient λ to 1.

For SCVI, VELOVI, and FlatVI, we use the default training parameters provided by the corresponding implementations, except for the maximum number of epochs, which is set to 400. Early stopping is enabled.

All available cells are used to train each latent representation. No train/validation split is applied at this stage. The purpose of this step is to obtain the latent representations used in the subsequent scDiffEq experiments.

3.4.2 Decoding cells for evaluation

In order to compare the decoded outputs of the different representations, we define a common post-processing procedure for the gene-expression profiles used in the evaluation. Since some decoder outputs may contain small negative values, these values are set to zero before normalization. This ensures that the resulting profiles can be interpreted as non-negative expression profiles.

For SCVI and FlatVI, we use the *px_scale* output provided by the models. The resulting expression profiles are non-negative and normalized so that the expression values of each cell sum to one.

For VELOVI, we use the expression output provided by the decoder. Negative values are set to zero, after which each cell is normalized so that its expression values sum to one.

For PCA, the predicted latent state is mapped back to gene-expression space using the inverse PCA transformation. Negative values in the reconstructed expression profile are set to zero, after which the profile is normalized to sum to one.

This post-processing procedure is applied before computing the optimal transport distances in gene-expression space.

3.5 Future cell-state prediction with scDiffEq

3.5.1 Input and output populations

We train and evaluate the scDiffEq model at the clone level. Formally, a clone corresponds to a group of cells observed at different time points (t_2 , t_4 , and t_6). We provide the initial cells of each clone as input to the model and compare the network outputs with the final cells belonging to the same clone.

Unless otherwise stated, we generate 32 cells per clone during both training and evaluation. These simulated cells are compared with the target cells observed at t_4 and t_6 . As previously explained, we split the dataset at the clone level to prevent data leakage.

3.5.2 Neural SDE architecture

The neural SDE network is composed of a drift network and a diffusion network. Both networks share the same architecture:

Drift/Diffusion network architecture
Input (latent dimension)
Dense layer (latent dimension, hidden dimension)
SiLU activation function
Dense layer (hidden dimension, hidden dimension)
SiLU activation function
Dense layer (hidden dimension, latent dimension)

The latent dimension is determined by the latent representation being studied (10, 20, 50, 100, and 200). We systematically investigate three values for the hidden dimension: 64, 128, and 256.

3.5.3 Future-state simulation

In order to generate future cells from the initial cells at t_0 , we use the Euler–Maruyama approximation:

$$z(t + \Delta t) = z(t) + f(z(t), t)\Delta t + g(z(t), t)\sqrt{\Delta t}\epsilon,$$

with $\Delta t = 0.05$. This corresponds to $2/0.05 = 40$ integration steps to generate cells from t_0 to t_4 , and $4/0.05 = 80$ integration steps to generate cells from t_0 to t_6 . ϵ is sampled from a standard normal distribution.

The final t_6 simulated cells are generated by continuing the simulation from the simulated cells at t_4 .

3.5.4 Training Objective

We denote the OT score between simulated cells at time t_i and measured cells at time t_j as OT_{t_i, t_j} . The objective function is defined as:

$$\mathcal{L} = \frac{1}{2}[OT_{t_4, t_4} + OT_{t_6, t_6}]$$

We use GeomLoss [9], a Python package, to solve the OT problems using the Sinkhorn algorithm. Sinkhorn is an entropy-regularized OT solver. It is widely used for solving OT problems because it is often more stable than solving the unregularized problem. We use the Euclidean distance as the cost function and set the blur parameter to 0.05.

We deviate from the scDiffEq implementation by not regularizing the objective function using a target drift-to-diffusion ratio. We made this choice because we did not find a justification for this regularization in the original scDiffEq paper.

3.5.5 Training procedure

We train the scDiffEq model at the clone level. Each batch contains 128 clones. We use Adam with a learning rate of 10^{-3} as the optimizer. The maximum number of epochs is set to 200. We use a PyTorch learning-rate scheduler to reduce the learning rate when the validation loss plateaus. Early stopping is also enabled.

The remaining implementation details are provided in A.

3.6 Benchmark configuration

We train 180 scDiffEq models corresponding to the following benchmark configuration:

Benchmark configuration	
method	PCA, FlatVI, VELOVI, SCVI
latent dimension	10, 20, 50, 100, 200
hidden dimension	64, 128, 256
seed	0, 1, 2

Importantly, every configuration is trained using exactly the same scDiffEq framework to ensure a fair comparison.

3.7 Evaluation

3.7.1 reconstruction quality evaluation

Since the evaluated models do not share the same probabilistic framework, it is not possible to directly compare their specific reconstruction errors. Instead, we provide raw counts as input to scVI, veloVI, flatVI models and normalized counts as input to PCA model. We normalized these inputs if needed and compare them to the normalized decoder outputs with the 3 distances defined in the optimal transport cost functions section.

3.7.2 Latent and gene-expression space evaluation

To evaluate a single scDiffEq model, we compute the mean OT score over the validation clones. The validation clones are determined by the corresponding train/validation split. We compute this score in both latent space and gene-expression space.

3.7.3 Baseline normalization

In order to fairly compare OT scores across different latent geometries, we introduce a baseline normalization. We define

$$\frac{OT_{t_j, t_j}}{OT_{t_0, t_j}}$$

as the normalized OT score at time t_j , where OT_{t_0, t_j} is the OT score computed between the initial measured cells and the final measured cells at time t_j .

This score provides an intuitive interpretation of model performance. A value of 1 corresponds to performance equivalent to using the initial state as the prediction. Values greater than 1 indicate worse performance, whereas values lower than 1 indicate better performance.

3.7.4 Optimal transport cost functions

To prevent the results from being influenced by the choice of cost function in the optimal transport problem, we use three different cost functions for the gene-expression-space evaluation:

- Euclidean L_2 distance.
- Cosine distance:

$$d_{\cos}(x, y) = 1 - \frac{x^\top y}{\|x\| \|y\|}.$$

- Hellinger distance:

$$d_H(x, y) = \sum_i (\sqrt{x_i} - \sqrt{y_i})^2.$$

The Hellinger distance is particularly appropriate because we are working with decoder outputs that are non-negative and normalized to sum to 1. In this representation, the data lie on the probability simplex. The Hellinger transformation maps the simplex to a portion of a hypersphere, making Euclidean geometry more natural in the transformed space.

3.8 Computational environment

All experiments were implemented in Python. The benchmark relied primarily on PyTorch, scvi-tools 1.4.2 and scanpy. The experiments were performed on linux ubuntu 24.04. The processor is an AMD Ryzen 5 8400F. 16Go of RAM was available. Nvidia RTX 5060 8Go of VRAM was used for model training and evaluation.

4 Results

4.1 Reconstruction Quality

We first evaluated the ability of the four latent representation models to reconstruct the observed gene-expression profiles. Reconstruction quality was assessed using three complementary distances: L2, cosine, and Hellinger-style distance. For each representation, the encoder and decoder were applied to the complete dataset, and the

Method	L2	Cosine	Hellinger
PCA	0.1758 (20)	0.2384 (20)	0.4044 (200)
SCVI	0.0756 (20)	0.1842 (50)	0.5418 (20)
VELOVI	0.3104 (50)	0.7757 (50)	1.3404 (50)
FlatVI	0.1354 (200)	0.5040 (200)	0.9557 (10)

Table 1: Best reconstruction performance obtained for each latent representation and distance metric. The latent dimension corresponding to the best result is reported in parentheses. Lower values indicate better reconstruction quality.

reconstructed expression profiles were compared with the corresponding observed cells after the same per-cell normalization.

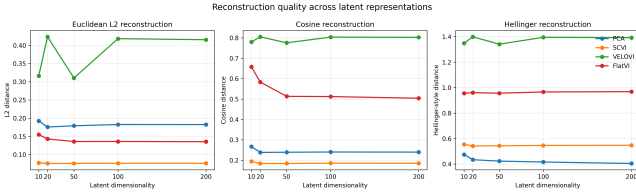


Figure 1: Reconstruction quality of the four latent representations across latent dimensions. The three panels report the mean L2, cosine, and Hellinger distances between the original and reconstructed gene-expression profiles. Lower values indicate better reconstruction quality.

Overall, SCVI achieved the lowest L2 and cosine distances across the tested latent dimensionalities. Its reconstruction performance remained relatively stable as the latent dimensionality increased, with L2 distances around 0.076 and cosine distances around 0.18–0.19.

PCA showed intermediate L2 and cosine distances, while its Hellinger-style distance was lower than that of SCVI. In particular, the Hellinger-style distance decreased from 0.475 at 10 latent dimensions to 0.404 at 200 dimensions. The L2 and cosine distances, however, showed little improvement beyond 20 latent dimensions.

FlatVI displayed an improvement in cosine distance with increasing latent dimensionality, decreasing from 0.658 at 10 dimensions to 0.504 at 200 dimensions. Its L2 distance also decreased, although less markedly. In contrast, its Hellinger-style distance remained relatively stable across latent dimensionalities.

VELOVI consistently showed the largest reconstruction distances across the three metrics. Its L2 distance ranged from 0.31 to 0.42, while its cosine and Hellinger-style distances remained substantially higher than those observed for the other representations.

These results indicate that reconstruction quality depends both on the latent representation and on the distance used for evaluation. In particular, SCVI provides the closest reconstructions according to L2 and cosine distance, whereas PCA performs best according to the Hellinger-style metric.

Detailed results are provided in [B.1](#).

Method	t_4	t_6
PCA	0.556 ± 0.012 (20)	0.529 ± 0.040 (20)
SCVI	0.491 ± 0.015 (20)	0.432 ± 0.009 (10)
VELOVI	0.549 ± 0.011 (100)	0.515 ± 0.013 (50)
FLATVI	0.534 ± 0.021 (20)	0.548 ± 0.043 (20)

Table 2: Future cell-state prediction performance in latent space. Values correspond to the mean normalized optimal transport (OT) score $\pm$ standard deviation across hidden dimensions and random seeds. The latent dimension yielding the lowest mean normalized OT score is reported in parentheses. Lower values indicate better performance.

4.2 Future cell-state prediction in latent space

We first investigate whether the choice of latent representation affects the performance of the scDiffEq models. The columns t_4 and t_6 report the normalized OT score at t_4 and t_6 , respectively, evaluated on the validation dataset.

Overall, SCVI achieves the lowest normalized OT score at both t_4 and t_6 . PCA, VELOVI, and FLATVI obtain slightly higher scores, although the differences between methods are relatively small compared with the variability across experimental configurations.

Importantly, all normalized OT scores are substantially below 1, ranging from 0.43 to 0.56. Since a value of 1 corresponds to the OT cost obtained when using the initial t_2 population as the prediction, these results indicate that all four latent representations allow scDiffEq to generate populations that are substantially closer to the target populations than the initial-state baseline.

The relatively similar performance across representations suggests that the choice of latent space does not strongly affect future-state prediction when the evaluation is performed directly in the latent space. However, this conclusion may depend on the representation space itself, as the latent spaces have different geometries and are therefore not necessarily directly comparable.

Detailed results across latent and hidden dimensions are provided in [B.2](#).

4.3 Future cell-state prediction in gene-expression space

We next evaluate the predicted populations in gene-expression space. For each method, we compute the normalized OT score using three different cost functions: L2, cosine, and Hellinger distances. The results are reported separately for t_4 and t_6 .

In gene-expression space, a clearer difference between the representation methods emerges. PCA and SCVI generally achieve lower normalized OT scores than VELOVI and FLATVI across the three cost functions and at both time points.

At t_4 , PCA achieves the lowest L2-based normalized OT score, whereas SCVI performs best according to the cosine and Hellinger-based metrics. A similar pattern is

Method	L2	Cosine	Hellinger
PCA	0.7688 $\pm$ 0.0095 (10)	0.8611 $\pm$ 0.0078 (10)	0.8775 $\pm$ 0.0059 (50)
SCVI	0.8227 $\pm$ 0.0046 (100)	0.7267 $\pm$ 0.0076 (10)	0.8653 $\pm$ 0.0020 (10)
VELOVI	1.0142 $\pm$ 0.0050 (200)	0.9581 $\pm$ 0.0016 (10)	0.9902 $\pm$ 0.0007 (10)
FLATVI	0.9716 $\pm$ 0.0032 (50)	0.9612 $\pm$ 0.0066 (50)	0.9775 $\pm$ 0.0010 (20)

Table 3: Future cell-state prediction performance in gene-expression space at t_4 . Values correspond to the mean normalized OT score $\pm$ standard deviation across random seeds. The best latent dimension and hidden dimension are reported in parentheses. Lower values indicate better performance.

Method	L2	Cosine	Hellinger
PCA	0.6921 $\pm$ 0.0432 (10)	0.7750 $\pm$ 0.0051 (10)	0.8076 $\pm$ 0.0055 (50)
SCVI	0.7720 $\pm$ 0.0136 (100)	0.6243 $\pm$ 0.0078 (10)	0.7903 $\pm$ 0.0087 (10)
VELOVI	0.9919 $\pm$ 0.0125 (10)	0.9241 $\pm$ 0.0068 (10)	0.9747 $\pm$ 0.0015 (10)
FLATVI	0.9644 $\pm$ 0.0053 (50)	0.9263 $\pm$ 0.0145 (50)	0.9589 $\pm$ 0.0036 (50)

Table 4: Future cell-state prediction performance in gene-expression space at t_6 . Values correspond to the mean normalized OT score $\pm$ standard deviation across random seeds. The best latent dimension and hidden dimension are reported in parentheses. Lower values indicate better performance.

observed at t_6 , with PCA obtaining the lowest L2 score and SCVI achieving the best cosine and Hellinger scores.

The difference between the methods is particularly apparent when considering the baseline value of 1. PCA and SCVI consistently obtain normalized OT scores below 1, indicating an improvement over using the initial t_2 population as the prediction. In contrast, VELOVI and FLATVI generally obtain scores close to 1, with VELOVI even slightly exceeding 1 for the L2 metric at t_4 . This suggests that, for these configurations, their predicted populations provide little or no improvement over the initial-state baseline when evaluated in gene-expression space.

Importantly, this qualitative pattern is consistent across the three cost functions. Although the absolute OT scores differ between L2, cosine, and Hellinger distances, PCA and SCVI tend to outperform VELOVI and FLATVI regardless of the chosen metric. This suggests that the observed difference between representation methods is not solely driven by the choice of cost function.

Overall, these results highlight a difference between evaluating future-state predictions in the latent space and in gene-expression space. While the four representations show relatively similar performance when evaluated in their respective latent spaces, a larger separation between methods emerges when the predictions are decoded and evaluated in the common gene-expression space.

Detailed results are provided in [B.3](#).

4.4 Relationship between reconstruction quality and future-state prediction

We next investigated whether reconstruction quality was associated with future cell-state prediction performance. Across the 20 representation configurations, reconstruction error showed a positive association with normalized

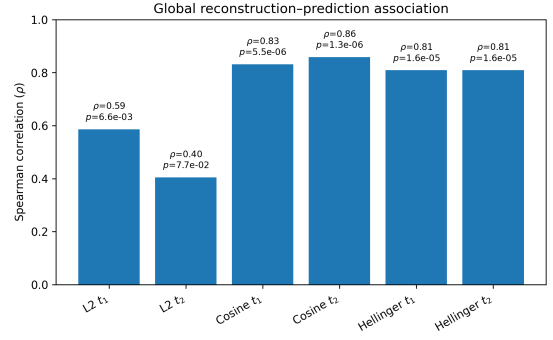


Figure 2: Global association between reconstruction error and future prediction error. Spearman correlations were computed across the 20 representation configurations (four methods and five latent dimensions). All global correlations are positive, with the strongest associations observed for the cosine and Hellinger distances.

gene-expression-space OT. This association was moderate for the L2 metric and substantially stronger for cosine and Hellinger distances. The relationship was observed at both prediction time points and was particularly strong for Hellinger and cosine distances. However, when correlations were computed separately within each representation method, the relationship became heterogeneous. This suggests that the strong global association is at least partly driven by differences between representation methods rather than by a universal relationship between reconstruction quality and predictive performance.

Detailed results are provided in [B.4](#).

5 Discussion

5.1 Main Results

Evaluating future gene-expression prediction across different latent-space representations is challenging. Many hyperparameters are involved in both the scDiffEq and latent representation models. Moreover, despite our efforts to make the benchmark as rigorous as possible, the evaluation metrics and normalization procedures remain partly dependent on methodological choices.

Nevertheless, our results show that PCA and scVI latent representations may perform better than FlatVI and VELOVI for future cell-state prediction in gene-expression space. One possible explanation is that PCA and scVI provide better reconstructions of gene-expression profiles, which may preserve information that is useful for future-state prediction. However, this interpretation should be treated with caution, since the relationship between reconstruction quality and future prediction performance is not consistently observed when considering each latent representation method separately. This suggests that the strong correlation observed across all 20 representations may be partly driven by systematic differences between representation methods rather than by a universal relationship between reconstruction quality and future prediction.

More generally, our results suggest that the choice

of latent representation can affect future cell-state prediction performance. Although the underlying reasons remain unclear, reconstruction metrics such as cosine and Hellinger distances show a strong association with future prediction performance when comparing different latent representations. These metrics may therefore provide useful information for estimating the relative predictive performance of different representations before performing the full future-state prediction benchmark.

Overall, these results suggest that the limitations of future cell-state prediction may be partly related to the ability of the latent representation and its decoder to preserve gene-expression information. However, this hypothesis should be interpreted cautiously. The four representation methods rely on substantially different probabilistic and reconstruction frameworks, making it difficult to completely normalize their behaviour and ensure a perfectly fair comparison. In addition, reconstruction quality alone does not appear to fully explain the differences in future-state prediction performance within individual representation families.

5.2 Limitations and Perspectives

Several limitations were identified throughout this study.

- Single dataset.** The benchmark was performed on a single dataset. Although the LARRY lineage-tracing dataset provides an appropriate setting for evaluating future cell-state prediction, the generality of our conclusions remains uncertain. Applying the benchmark to additional datasets, ideally covering different biological systems and experimental conditions, would strengthen the robustness and generalizability of the results.
- Dependence on the scDiffEq prediction framework.** We assume that future cell-state prediction can be adequately represented by the scDiffEq framework and its neural SDE formulation. Other future-state prediction approaches could produce different results. Comparing scDiffEq with alternative dynamical models would therefore help determine whether the observed differences between latent representations are specific to this framework or reflect more general properties of the representations.
- Differences between latent representation models.** The benchmarked latent representation methods rely on substantially different probabilistic assumptions and training objectives. PCA is a linear deterministic method, whereas scVI, VELOVI, and FlatVI rely on different probabilistic generative models and regularization schemes. Consequently, defining a completely fair comparison between these representations is challenging. In particular, their latent spaces may have different geometries and their decoders may preserve different aspects of gene-expression variation. The normalization procedures used in this study reduce some of these differences, but cannot guarantee that no representation method benefits from the evaluation procedure.
- Choice of evaluation metrics.** The benchmark relies on a limited set of distance metrics, namely L2, cosine, and Hellinger distances. Each metric emphasizes different aspects of the gene-expression distribution. Therefore, the ranking of the representation methods may depend partly on the selected metric. Additional distributional or biologically informed metrics could provide complementary information.
- Latent-space comparability.** Latent representations are not necessarily directly comparable in terms of Euclidean geometry. The same numerical distance in two different latent spaces does not necessarily correspond to the same biological difference. This limitation is particularly relevant when evaluating future predictions directly in latent space. Gene-expression-space evaluation partially addresses this issue by bringing predictions back into a common representation, but it introduces additional dependence on the decoder of each representation model.
- Reconstruction versus dynamical information.** Reconstruction quality measures how well a latent representation preserves the current gene-expression state, but it does not directly measure whether the representation preserves information relevant to future cellular dynamics. The absence of a consistent within-method relationship between reconstruction and prediction performance suggests that a representation can preserve current-state information without necessarily providing the most suitable geometry for modelling future dynamics.

These limitations also motivate several directions for further work. First, the benchmark could be extended to additional datasets and prediction frameworks to assess the generality of the observed trends. Second, alternative evaluation metrics could be investigated, particularly metrics that capture biologically meaningful differences between predicted and observed cell states. Finally, it would be valuable to investigate whether latent representations can be explicitly optimized for future-state prediction rather than reconstruction alone. Such approaches could potentially combine reconstruction quality with properties of the latent geometry that are better suited for modelling cellular dynamics.

References

- [1] Romain Lopez, Jeffrey Regier, Michael B. Cole, Michael I. Jordan, and Nir Yosef. Deep generative modeling for single-cell transcriptomics. *Nature Methods*, 15(12):1053–1058, 2018.
- [2] Grace H. T. Yeo, Joshua Hwang, Wouter Saelens, and David K. Gifford. Generative modeling of single-cell time series with prescient enables prediction of cell trajectories with interventions. *Nature Communications*, 12(1):3222, May 2021.
- [3] Michael E. Vinyard et al. Learning cell dynamics with neural differential equations. *Nature Machine Intelligence*, 7:1969–1984, 2025.
- [4] Ian T. Jolliffe. *Principal Component Analysis*. Springer-Verlag, New York, second edition, 2002.
- [5] Adam Gayoso, Philipp Weiler, Mohammad Lotfollahi, Amit Frishberg, Chenling Su, Ksenia Klimovskaia, Chenling Xu, Nir Yosef, Fabian J Theis, and Volker Bergen. Deep generative modeling of transcriptional dynamics for rna velocity analysis in single cells. *Nature Methods*, 21(1):64–73, 2024.
- [6] Leon Hetzel Stephan Günnemann Fabian J. Theis Alessandro Palma, Sergei Rybakov. Enforcing latent euclidean geometry in single-cell vaes for manifold interpolation. *arXiv preprint arXiv:2507.11789*, 2025.
- [7] Geoffrey Schiebinger, Jian Shu, Marcin Tabaka, Brian Cleary, Vasanth Subramanian, Justin Solomon, Jett Gould, Siyan Liu, Stacie Lin, Peter Berube, Lia Lee, Jenny Zhao, Jialin Zhang, Thomas Dietmeister, Aviv Regev, and Eric S Lander. Optimal-transport analysis of single-cell gene expression identifies developmental trajectories in reprogramming. *Cell*, 176(4):928–943, 2019.
- [8] Caleb Weinreb, Alejo Rodriguez-Fraticelli, Mandinema H Chou, Ruhama Weinreb, Wang Wang, and Alon M Klein. Lineage tracing on transcriptional landscapes links state to fate during differentiation. *Science*, 367(6479):eaaw3381, 2020.
- [9] Jean Feydy, Thibault Séjourné, François-Xavier Vialard, Shun-ichi Amari, Alain Trounev, and Gabriel Peyré. Interpolating between optimal transport and mmd using sinkhorn divergences. In *Proceedings of the Twenty-Second International Conference on Artificial Intelligence and Statistics (AISTATS)*, volume 89 of *Proceedings of Machine Learning Research*, pages 2681–2690. PMLR, 2019.

Appendix A Scdifreq training implementation

We provide in this section the implementations details used to train all the scdifreq models in our benchmark

A.1 reduceLROnPlateau parameters

torch reduceLROnPlateau parameters	
mode	min
factor	0.5
patience	10
threshold	1e-3
thresholdmodel	rel
cooldown	0
minlr	1e-6

A.2 Early stopping parameters

Early stopping parameters	
patience	20
min relative improvement	5e-3

Appendix B Results

B.1 Reconstruction Quality

We provide the detailed reconstruction quality results:

Method	Latent dim.	L2	Cosine	Hellinger
PCA	10	0.1925 $\pm$ 0.1033	0.2666 $\pm$ 0.1456	0.4748 $\pm$ 0.1799
PCA	20	0.1758 $\pm$ 0.1016	0.2384 $\pm$ 0.1317	0.4341 $\pm$ 0.1704
PCA	50	0.1791 $\pm$ 0.1024	0.2393 $\pm$ 0.1325	0.4231 $\pm$ 0.1640
PCA	100	0.1826 $\pm$ 0.1066	0.2404 $\pm$ 0.1340	0.4161 $\pm$ 0.1611
PCA	200	0.1825 $\pm$ 0.1064	0.2400 $\pm$ 0.1346	0.4044 $\pm$ 0.1547
SCVI	10	0.0773 $\pm$ 0.0314	0.1941 $\pm$ 0.1347	0.5541 $\pm$ 0.2419
SCVI	20	0.0756 $\pm$ 0.0308	0.1845 $\pm$ 0.1303	0.5418 $\pm$ 0.2423
SCVI	50	0.0759 $\pm$ 0.0316	0.1842 $\pm$ 0.1301	0.5434 $\pm$ 0.2427
SCVI	100	0.0760 $\pm$ 0.0314	0.1857 $\pm$ 0.1306	0.5460 $\pm$ 0.2432
SCVI	200	0.0761 $\pm$ 0.0315	0.1853 $\pm$ 0.1305	0.5475 $\pm$ 0.2425
VELOVI	10	0.3161 $\pm$ 0.1808	0.7798 $\pm$ 0.2651	1.3479 $\pm$ 0.2869
VELOVI	20	0.4235 $\pm$ 0.2136	0.8051 $\pm$ 0.2738	1.3984 $\pm$ 0.3193
VELOVI	50	0.3104 $\pm$ 0.1759	0.7757 $\pm$ 0.2644	1.3404 $\pm$ 0.2853
VELOVI	100	0.4184 $\pm$ 0.2125	0.8036 $\pm$ 0.2726	1.3944 $\pm$ 0.3154
VELOVI	200	0.4155 $\pm$ 0.2108	0.8024 $\pm$ 0.2726	1.3915 $\pm$ 0.3147
FlatVI	10	0.1551 $\pm$ 0.0744	0.6581 $\pm$ 0.1022	0.9557 $\pm$ 0.2136
FlatVI	20	0.1429 $\pm$ 0.0748	0.5834 $\pm$ 0.1218	0.9605 $\pm$ 0.2157
FlatVI	50	0.1359 $\pm$ 0.0760	0.5133 $\pm$ 0.1366	0.9560 $\pm$ 0.2147
FlatVI	100	0.1361 $\pm$ 0.0759	0.5121 $\pm$ 0.1295	0.9656 $\pm$ 0.2093
FlatVI	200	0.1354 $\pm$ 0.0758	0.5040 $\pm$ 0.1354	0.9680 $\pm$ 0.2102

Table 5: Reconstruction performance for each method and latent dimension. Values are reported as mean $\pm$ standard deviation. Lower values indicate better reconstruction quality.

B.2 Future cell-state prediction in latent space

Table 6: Complete latent-space benchmark results across latent dimensions. Values correspond to the mean normalized optimal transport (OT) score $\pm$ standard deviation across hidden dimensions and random seeds.

Method	Latent dim.	t_4 mean	t_4 std	t_6 mean	t_6 std
FLATVI	10	0.5790	0.0182	0.5907	0.0306
FLATVI	10	0.5714	0.0209	0.5878	0.0451
FLATVI	10	0.5649	0.0292	0.5817	0.0442

Method	Latent dim.	t_4 mean	t_4 std	t_6 mean	t_6 std
FLATVI	20	0.5440	0.0218	0.5583	0.0482
FLATVI	20	0.5273	0.0282	0.5424	0.0624
FLATVI	20	0.5299	0.0174	0.5423	0.0285
FLATVI	50	0.6018	0.0352	0.6059	0.0403
FLATVI	50	0.5911	0.0459	0.6081	0.0404
FLATVI	50	0.5894	0.0503	0.6325	0.0404
FLATVI	100	0.6898	0.0607	0.6596	0.0782
FLATVI	100	0.6458	0.0482	0.6350	0.0642
FLATVI	100	0.6454	0.0622	0.6318	0.0720
FLATVI	200	0.6669	0.0461	0.6518	0.0272
FLATVI	200	0.6238	0.0353	0.5902	0.0461
FLATVI	200	0.6292	0.0309	0.6250	0.0450
PCA	10	0.6561	0.0426	0.6311	0.1049
PCA	10	0.6427	0.0344	0.6032	0.1084
PCA	10	0.6446	0.0236	0.6061	0.1211
PCA	20	0.5541	0.0069	0.5217	0.0459
PCA	20	0.5657	0.0119	0.5389	0.0375
PCA	20	0.5471	0.0101	0.5251	0.0523
PCA	50	0.5765	0.0120	0.5394	0.0497
PCA	50	0.5670	0.0073	0.5264	0.0374
PCA	50	0.5582	0.0010	0.5251	0.0381
PCA	100	0.6425	0.0199	0.5906	0.0291
PCA	100	0.5752	0.0126	0.5273	0.0714
PCA	100	0.5572	0.0041	0.5092	0.0386
PCA	200	0.7182	0.0224	0.6603	0.0478
PCA	200	0.6307	0.0068	0.5800	0.0543
PCA	200	0.5683	0.0055	0.5133	0.0518
SCVI	10	0.5298	0.0079	0.4341	0.0069
SCVI	10	0.5265	0.0116	0.4343	0.0085
SCVI	10	0.5192	0.0094	0.4271	0.0128
SCVI	20	0.4955	0.0145	0.4486	0.0047
SCVI	20	0.4884	0.0189	0.4396	0.0065
SCVI	20	0.4889	0.0175	0.4399	0.0053
SCVI	50	0.5075	0.0124	0.4724	0.0096
SCVI	50	0.4977	0.0167	0.4623	0.0070
SCVI	50	0.4936	0.0174	0.4581	0.0070
SCVI	100	0.5066	0.0141	0.4716	0.0060
SCVI	100	0.4997	0.0149	0.4620	0.0015
SCVI	100	0.4927	0.0191	0.4561	0.0096
SCVI	200	0.5298	0.0145	0.4916	0.0050
SCVI	200	0.5161	0.0199	0.4773	0.0072
SCVI	200	0.5120	0.0199	0.4707	0.0063
VELOVI	10	0.5867	0.0011	0.5716	0.0162
VELOVI	10	0.5821	0.0083	0.5655	0.0178
VELOVI	10	0.5848	0.0050	0.5730	0.0157
VELOVI	20	0.5715	0.0172	0.5704	0.0114
VELOVI	20	0.5662	0.0222	0.5616	0.0175
VELOVI	20	0.5586	0.0216	0.5476	0.0026
VELOVI	50	0.5565	0.0086	0.5202	0.0153
VELOVI	50	0.5537	0.0009	0.5209	0.0083
VELOVI	50	0.5388	0.0010	0.5042	0.0108
VELOVI	100	0.5583	0.0049	0.5389	0.0164
VELOVI	100	0.5459	0.0131	0.5272	0.0213
VELOVI	100	0.5418	0.0082	0.5252	0.0179
VELOVI	200	0.5932	0.0159	0.5820	0.0078
VELOVI	200	0.5836	0.0107	0.5686	0.0084
VELOVI	200	0.5690	0.0094	0.5570	0.0101

B.3 Future cell-state prediction in gene-expression space

Table 7: Complete gene-expression-space benchmark results at t_4 . Values correspond to the mean $\pm$ standard deviation of the normalized optimal transport (OT) score across random seeds. Results are reported for each method, latent dimension, and hidden dimension. Lower values indicate better performance.

Method	Latent dim.	Hidden dim.	L2	Cosine	Hellinger
FLATVI	10	64	1.0178 ± 0.0044	1.0098 ± 0.0032	0.9799 ± 0.0005
FLATVI	10	128	1.0145 ± 0.0050	1.0070 ± 0.0030	0.9791 ± 0.0013
FLATVI	10	256	1.0127 ± 0.0046	1.0060 ± 0.0023	0.9784 ± 0.0015
FLATVI	20	64	0.9887 ± 0.0014	0.9880 ± 0.0015	0.9789 ± 0.0014
FLATVI	20	128	0.9869 ± 0.0017	0.9850 ± 0.0028	0.9782 ± 0.0008
FLATVI	20	256	0.9850 ± 0.0014	0.9842 ± 0.0024	0.9775 ± 0.0010
FLATVI	50	64	0.9749 ± 0.0035	0.9662 ± 0.0058	0.9790 ± 0.0017
FLATVI	50	128	0.9718 ± 0.0033	0.9623 ± 0.0060	0.9781 ± 0.0005
FLATVI	50	256	0.9716 ± 0.0032	0.9612 ± 0.0066	0.9782 ± 0.0017
FLATVI	100	64	0.9796 ± 0.0029	0.9784 ± 0.0054	0.9834 ± 0.0019
FLATVI	100	128	0.9762 ± 0.0017	0.9707 ± 0.0038	0.9815 ± 0.0006
FLATVI	100	256	0.9774 ± 0.0025	0.9709 ± 0.0052	0.9825 ± 0.0017
FLATVI	200	64	0.9791 ± 0.0018	0.9764 ± 0.0024	0.9827 ± 0.0007
FLATVI	200	128	0.9779 ± 0.0018	0.9702 ± 0.0039	0.9820 ± 0.0013
FLATVI	200	256	0.9766 ± 0.0031	0.9708 ± 0.0030	0.9815 ± 0.0009
PCA	10	64	0.7729 ± 0.0076	0.8614 ± 0.0006	0.8913 ± 0.0058
PCA	10	128	0.7754 ± 0.0058	0.8633 ± 0.0021	0.8891 ± 0.0040
PCA	10	256	0.7688 ± 0.0095	0.8611 ± 0.0078	0.8880 ± 0.0030
PCA	20	64	0.8451 ± 0.0167	0.8951 ± 0.0065	0.8832 ± 0.0071
PCA	20	128	0.8523 ± 0.0097	0.8988 ± 0.0066	0.8832 ± 0.0065
PCA	20	256	0.8837 ± 0.0048	0.9115 ± 0.0035	0.8836 ± 0.0049
PCA	50	64	0.8339 ± 0.0286	0.8850 ± 0.0138	0.8786 ± 0.0046
PCA	50	128	0.8346 ± 0.0214	0.8841 ± 0.0107	0.8775 ± 0.0059
PCA	50	256	0.8533 ± 0.0188	0.8904 ± 0.0079	0.8781 ± 0.0036
PCA	100	64	0.8072 ± 0.0224	0.8731 ± 0.0037	0.8822 ± 0.0058
PCA	100	128	0.8342 ± 0.0294	0.8777 ± 0.0121	0.8783 ± 0.0070
PCA	100	256	0.8419 ± 0.0219	0.8799 ± 0.0058	0.8781 ± 0.0088
PCA	200	64	0.8046 ± 0.0197	0.8763 ± 0.0052	0.8839 ± 0.0083
PCA	200	128	0.8583 ± 0.0411	0.8906 ± 0.0140	0.8808 ± 0.0124
PCA	200	256	0.8823 ± 0.0365	0.8919 ± 0.0181	0.8795 ± 0.0124
SCVI	10	64	0.8266 ± 0.0068	0.7292 ± 0.0099	0.8675 ± 0.0010
SCVI	10	128	0.8311 ± 0.0106	0.7313 ± 0.0061	0.8682 ± 0.0027
SCVI	10	256	0.8284 ± 0.0121	0.7267 ± 0.0076	0.8653 ± 0.0020
SCVI	20	64	0.8302 ± 0.0072	0.7315 ± 0.0020	0.8713 ± 0.0027
SCVI	20	128	0.8273 ± 0.0069	0.7280 ± 0.0051	0.8699 ± 0.0028
SCVI	20	256	0.8245 ± 0.0088	0.7269 ± 0.0033	0.8687 ± 0.0030
SCVI	50	64	0.8338 ± 0.0084	0.7373 ± 0.0105	0.8764 ± 0.0027
SCVI	50	128	0.8265 ± 0.0083	0.7279 ± 0.0064	0.8713 ± 0.0013
SCVI	50	256	0.8311 ± 0.0101	0.7312 ± 0.0044	0.8709 ± 0.0012
SCVI	100	64	0.8338 ± 0.0090	0.7414 ± 0.0009	0.8793 ± 0.0027
SCVI	100	128	0.8259 ± 0.0101	0.7332 ± 0.0059	0.8753 ± 0.0025
SCVI	100	256	0.8227 ± 0.0046	0.7271 ± 0.0038	0.8722 ± 0.0043
SCVI	200	64	0.8622 ± 0.0063	0.7555 ± 0.0067	0.8880 ± 0.0004
SCVI	200	128	0.8539 ± 0.0033	0.7457 ± 0.0048	0.8796 ± 0.0030
SCVI	200	256	0.8526 ± 0.0115	0.7394 ± 0.0024	0.8793 ± 0.0019
VELOVI	10	64	1.0204 ± 0.0091	0.9598 ± 0.0019	0.9905 ± 0.0008
VELOVI	10	128	1.0183 ± 0.0103	0.9581 ± 0.0016	0.9902 ± 0.0007
VELOVI	10	256	1.0259 ± 0.0159	0.9584 ± 0.0035	0.9904 ± 0.0008
VELOVI	20	64	1.0184 ± 0.0067	0.9716 ± 0.0044	0.9937 ± 0.0012
VELOVI	20	128	1.0201 ± 0.0098	0.9714 ± 0.0069	0.9932 ± 0.0015
VELOVI	20	256	1.0205 ± 0.0082	0.9723 ± 0.0057	0.9936 ± 0.0015
VELOVI	50	64	1.0324 ± 0.0150	0.9661 ± 0.0048	0.9932 ± 0.0013
VELOVI	50	128	1.0215 ± 0.0032	0.9637 ± 0.0052	0.9923 ± 0.0016

Method	Latent dim.	Hidden dim.	L2	Cosine	Hellinger
VELOVI	50	256	1.0336 ± 0.0085	0.9636 ± 0.0047	0.9919 ± 0.0019
VELOVI	100	64	1.0210 ± 0.0100	0.9742 ± 0.0038	0.9942 ± 0.0014
VELOVI	100	128	1.0236 ± 0.0054	0.9743 ± 0.0030	0.9938 ± 0.0008
VELOVI	100	256	1.0262 ± 0.0114	0.9733 ± 0.0040	0.9936 ± 0.0016
VELOVI	200	64	1.0239 ± 0.0050	0.9757 ± 0.0038	0.9941 ± 0.0008
VELOVI	200	128	1.0203 ± 0.0074	0.9736 ± 0.0048	0.9937 ± 0.0014
VELOVI	200	256	1.0142 ± 0.0050	0.9721 ± 0.0028	0.9933 ± 0.0004

Table 8: Complete gene-expression-space benchmark results at t_6 . Values correspond to the mean $\pm$ standard deviation of the normalized optimal transport (OT) score across random seeds. Results are reported for each method, latent dimension, and hidden dimension. Lower values indicate better performance.

Method	Latent dim.	Hidden dim.	L2	Cosine	Hellinger
FLATVI	10	64	1.0196 ± 0.0055	0.9899 ± 0.0061	0.9620 ± 0.0035
FLATVI	10	128	1.0173 ± 0.0044	0.9875 ± 0.0041	0.9616 ± 0.0041
FLATVI	10	256	1.0144 ± 0.0019	0.9863 ± 0.0046	0.9606 ± 0.0041
FLATVI	20	64	0.9836 ± 0.0038	0.9576 ± 0.0067	0.9623 ± 0.0037
FLATVI	20	128	0.9819 ± 0.0036	0.9536 ± 0.0087	0.9618 ± 0.0036
FLATVI	20	256	0.9813 ± 0.0014	0.9555 ± 0.0065	0.9606 ± 0.0032
FLATVI	50	64	0.9670 ± 0.0058	0.9269 ± 0.0125	0.9603 ± 0.0048
FLATVI	50	128	0.9644 ± 0.0053	0.9264 ± 0.0145	0.9589 ± 0.0036
FLATVI	50	256	0.9654 ± 0.0064	0.9263 ± 0.0145	0.9595 ± 0.0045
FLATVI	100	64	0.9740 ± 0.0049	0.9413 ± 0.0084	0.9676 ± 0.0033
FLATVI	100	128	0.9702 ± 0.0038	0.9330 ± 0.0094	0.9649 ± 0.0038
FLATVI	100	256	0.9713 ± 0.0060	0.9345 ± 0.0118	0.9661 ± 0.0047
FLATVI	200	64	0.9728 ± 0.0034	0.9370 ± 0.0085	0.9668 ± 0.0036
FLATVI	200	128	0.9702 ± 0.0045	0.9299 ± 0.0080	0.9656 ± 0.0032
FLATVI	200	256	0.9719 ± 0.0030	0.9361 ± 0.0084	0.9663 ± 0.0029
PCA	10	64	0.6921 ± 0.0432	0.7750 ± 0.0051	0.8239 ± 0.0073
PCA	10	128	0.7150 ± 0.0134	0.7813 ± 0.0021	0.8235 ± 0.0089
PCA	10	256	0.7051 ± 0.0288	0.7760 ± 0.0085	0.8223 ± 0.0069
PCA	20	64	0.7622 ± 0.0129	0.7945 ± 0.0064	0.8121 ± 0.0046
PCA	20	128	0.7601 ± 0.0274	0.7998 ± 0.0208	0.8137 ± 0.0104
PCA	20	256	0.8081 ± 0.0216	0.8084 ± 0.0066	0.8148 ± 0.0048
PCA	50	64	0.7920 ± 0.0322	0.7979 ± 0.0149	0.8107 ± 0.0055
PCA	50	128	0.7357 ± 0.0261	0.7832 ± 0.0071	0.8076 ± 0.0055
PCA	50	256	0.7705 ± 0.0290	0.7900 ± 0.0096	0.8090 ± 0.0016
PCA	100	64	0.7607 ± 0.0468	0.7965 ± 0.0095	0.8165 ± 0.0061
PCA	100	128	0.7865 ± 0.0152	0.7909 ± 0.0091	0.8088 ± 0.0080
PCA	100	256	0.7828 ± 0.0346	0.7886 ± 0.0160	0.8094 ± 0.0070
PCA	200	64	0.7532 ± 0.0339	0.7955 ± 0.0110	0.8161 ± 0.0086
PCA	200	128	0.8659 ± 0.0885	0.8190 ± 0.0330	0.8158 ± 0.0199
PCA	200	256	0.8870 ± 0.0897	0.8100 ± 0.0438	0.8154 ± 0.0229
SCVI	10	64	0.7758 ± 0.0057	0.6273 ± 0.0073	0.7939 ± 0.0101
SCVI	10	128	0.7877 ± 0.0112	0.6373 ± 0.0052	0.7959 ± 0.0071
SCVI	10	256	0.7724 ± 0.0159	0.6243 ± 0.0078	0.7903 ± 0.0087
SCVI	20	64	0.7898 ± 0.0125	0.6397 ± 0.0072	0.8009 ± 0.0063
SCVI	20	128	0.7791 ± 0.0134	0.6284 ± 0.0125	0.7935 ± 0.0085
SCVI	20	256	0.7798 ± 0.0096	0.6305 ± 0.0080	0.7959 ± 0.0029
SCVI	50	64	0.7969 ± 0.0175	0.6491 ± 0.0022	0.8107 ± 0.0056
SCVI	50	128	0.7785 ± 0.0191	0.6325 ± 0.0230	0.7997 ± 0.0066
SCVI	50	256	0.7820 ± 0.0189	0.6328 ± 0.0126	0.7980 ± 0.0055
SCVI	100	64	0.7930 ± 0.0120	0.6530 ± 0.0181	0.8117 ± 0.0076
SCVI	100	128	0.7765 ± 0.0079	0.6333 ± 0.0091	0.8046 ± 0.0084
SCVI	100	256	0.7720 ± 0.0136	0.6275 ± 0.0162	0.7994 ± 0.0083
SCVI	200	64	0.8285 ± 0.0225	0.6751 ± 0.0048	0.8221 ± 0.0051
SCVI	200	128	0.8089 ± 0.0099	0.6523 ± 0.0145	0.8109 ± 0.0023
SCVI	200	256	0.7986 ± 0.0202	0.6399 ± 0.0132	0.8078 ± 0.0043

Method	Latent dim.	Hidden dim.	L2	Cosine	Hellinger
VELOVI	10	64	0.9933 ± 0.0071	0.9263 ± 0.0078	0.9755 ± 0.0016
VELOVI	10	128	0.9919 ± 0.0125	0.9254 ± 0.0070	0.9749 ± 0.0019
VELOVI	10	256	1.0007 ± 0.0134	0.9241 ± 0.0068	0.9747 ± 0.0015
VELOVI	20	64	1.0120 ± 0.0060	0.9553 ± 0.0080	0.9812 ± 0.0026
VELOVI	20	128	1.0081 ± 0.0082	0.9526 ± 0.0086	0.9805 ± 0.0023
VELOVI	20	256	1.0082 ± 0.0148	0.9531 ± 0.0095	0.9807 ± 0.0029
VELOVI	50	64	1.0048 ± 0.0107	0.9301 ± 0.0079	0.9767 ± 0.0006
VELOVI	50	128	0.9978 ± 0.0068	0.9290 ± 0.0094	0.9763 ± 0.0012
VELOVI	50	256	1.0011 ± 0.0156	0.9263 ± 0.0108	0.9751 ± 0.0019
VELOVI	100	64	1.0181 ± 0.0038	0.9558 ± 0.0057	0.9812 ± 0.0010
VELOVI	100	128	1.0151 ± 0.0089	0.9527 ± 0.0089	0.9808 ± 0.0028
VELOVI	100	256	1.0175 ± 0.0071	0.9538 ± 0.0050	0.9808 ± 0.0008
VELOVI	200	64	1.0089 ± 0.0085	0.9554 ± 0.0039	0.9808 ± 0.0009
VELOVI	200	128	1.0053 ± 0.0064	0.9526 ± 0.0079	0.9802 ± 0.0016
VELOVI	200	256	1.0097 ± 0.0099	0.9524 ± 0.0058	0.9804 ± 0.0020

B.4 Relationship between reconstruction quality and future-state prediction

Table 9: Pearson and Spearman correlations between reconstruction error and future prediction error. Global correlations were computed across the 20 representation configurations, whereas within-method correlations were computed across the five latent dimensions of each representation method.

Analysis	Method	Metric	Time	n	Pearson r	Pearson p	Spearman ρ	Spearman p
Global	All	L2	t_4	20	0.6407	2.34×10^{-3}	0.5865	6.57×10^{-3}
Within-method	FLATVI	L2	t_4	5	0.9865	1.88×10^{-3}	0.7000	1.88×10^{-1}
Within-method	PCA	L2	t_4	5	-0.9518	1.26×10^{-2}	-0.9000	3.74×10^{-2}
Within-method	SCVI	L2	t_4	5	-0.0611	9.22×10^{-1}	0.5000	3.91×10^{-1}
Within-method	VELOVI	L2	t_4	5	-0.6366	2.48×10^{-1}	-0.5000	3.91×10^{-1}
Global	All	L2	t_6	20	0.5932	5.83×10^{-3}	0.4045	7.69×10^{-2}
Within-method	FLATVI	L2	t_6	5	0.9857	2.05×10^{-3}	0.9000	3.74×10^{-2}
Within-method	PCA	L2	t_6	5	-0.5916	2.93×10^{-1}	-0.5000	3.91×10^{-1}
Within-method	SCVI	L2	t_6	5	-0.2196	7.23×10^{-1}	-0.3000	6.24×10^{-1}
Within-method	VELOVI	L2	t_6	5	0.8644	5.87×10^{-2}	0.8000	1.04×10^{-1}
Global	All	Cosine	t_4	20	0.8163	1.14×10^{-5}	0.8316	5.52×10^{-6}
Within-method	FLATVI	Cosine	t_4	5	0.9627	8.59×10^{-3}	0.7000	1.88×10^{-1}
Within-method	PCA	Cosine	t_4	5	-0.8260	8.48×10^{-2}	-1.0000	1.40×10^{-24}
Within-method	SCVI	Cosine	t_4	5	-0.3152	6.05×10^{-1}	0.1000	8.73×10^{-1}
Within-method	VELOVI	Cosine	t_4	5	0.8999	3.74×10^{-2}	0.6000	2.85×10^{-1}
Global	All	Cosine	t_6	20	0.8728	5.19×10^{-7}	0.8586	1.27×10^{-6}
Within-method	FLATVI	Cosine	t_6	5	0.9810	3.13×10^{-3}	0.7000	1.88×10^{-1}
Within-method	PCA	Cosine	t_6	5	-0.7880	1.13×10^{-1}	-0.5000	3.91×10^{-1}
Within-method	SCVI	Cosine	t_6	5	-0.4576	4.38×10^{-1}	-0.5000	3.91×10^{-1}
Within-method	VELOVI	Cosine	t_6	5	0.9820	2.90×10^{-3}	0.8000	1.04×10^{-1}
Global	All	Hellinger	t_4	20	0.9329	2.04×10^{-9}	0.8090	1.57×10^{-5}
Within-method	FLATVI	Hellinger	t_4	5	0.8591	6.21×10^{-2}	0.5000	3.91×10^{-1}
Within-method	PCA	Hellinger	t_4	5	0.8684	5.62×10^{-2}	0.6000	2.85×10^{-1}
Within-method	SCVI	Hellinger	t_4	5	-0.2046	7.41×10^{-1}	0.0000	1.00
Within-method	VELOVI	Hellinger	t_4	5	0.7943	1.08×10^{-1}	0.6000	2.85×10^{-1}
Global	All	Hellinger	t_6	20	0.9291	3.33×10^{-9}	0.8090	1.57×10^{-5}
Within-method	FLATVI	Hellinger	t_6	5	0.9423	1.65×10^{-2}	0.9000	3.74×10^{-2}
Within-method	PCA	Hellinger	t_6	5	0.7425	1.51×10^{-1}	0.3000	6.24×10^{-1}
Within-method	SCVI	Hellinger	t_6	5	-0.2254	7.16×10^{-1}	0.0000	1.00
Within-method	VELOVI	Hellinger	t_6	5	0.9733	5.22×10^{-3}	0.8000	1.04×10^{-1}

Appendix C Latent Space Metric Tensor Study

To further characterize the geometry of the latent representations, we computed the decoder-induced metric tensor for each representation model. We summarize the local geometry using the condition number of the metric tensor, defined as the ratio between its largest and smallest eigenvalues. A high condition number indicates that local distances in the latent space are strongly distorted when mapped to the corresponding representation space.

Let D denote the decoder, z a point in the latent space, and x the corresponding point in the representation space. We have

$$x = D(z). \quad (1)$$

For a small displacement δz in the latent space, the corresponding displacement in the representation space can be approximated using a first-order Taylor expansion:

$$\delta x = D(z + \delta z) - D(z) \approx J_D(z) \delta z, \quad (2)$$

where $J_D(z)$ denotes the Jacobian of the decoder. Therefore,

$$\|\delta x\|^2 = \delta z^\top J_D(z)^\top J_D(z) \delta z. \quad (3)$$

This defines the decoder-induced metric tensor

$$G_D(z) = J_D(z)^\top J_D(z), \quad (4)$$

such that

$$\|\delta x\|^2 = \delta z^\top G_D(z) \delta z. \quad (5)$$

For the PCA representation, the decoder is a linear transformation, $x = Wz$, where W contains the principal component directions. Since the PCA components are orthonormal, $W^\top W = I$, and therefore

$$G_D(z) = I. \quad (6)$$

Consequently, the PCA latent space has a constant Euclidean metric.

For probabilistic latent representation models, the decoder does not necessarily define a deterministic gene-expression state. Instead, the latent variable parameterizes a probability distribution over gene-expression profiles. Let $\theta(z)$ denote the parameters of the corresponding observation distribution:

$$p(x | z) = p(x | \theta(z)). \quad (7)$$

The Fisher information matrix in the parameter space is defined as

$$F_\theta(z) = \mathbb{E}_{x \sim p(x|\theta(z))} [\nabla_\theta \log p(x | \theta) \nabla_\theta \log p(x | \theta)^\top]. \quad (8)$$

The Fisher information induces a metric in the latent space through the Jacobian of the mapping from z to the distribution parameters:

$$G(z) = J_\theta(z)^\top F_\theta(z) J_\theta(z). \quad (9)$$

This formulation accounts for the probabilistic observation model associated with each representation method. Consequently, the Fisher information matrix is model-dependent and depends on the likelihood used by the corresponding model.

For the Negative Binomial likelihood used by SCVI and FlatVI, the Fisher information can be derived from (8). Denoting the distribution parameters by $\theta(z) = (\mu, \alpha)$, we write

$$F_\theta(z) = \mathcal{F}(\mu, \alpha), \quad (10)$$

with

$$\mathcal{F}(\mu, \alpha) = \begin{pmatrix} \frac{1}{\mu(1+\alpha\mu)} & 0 \\ 0 & \mathcal{I}_{\alpha\alpha} \end{pmatrix}, \quad (11)$$

$$\mathcal{I}_{\alpha\alpha} = \frac{1}{\alpha^4} \mathbb{E} \left[\psi' \left(\frac{1}{\alpha} \right) - \psi' \left(\frac{1}{\alpha} + X \right) \right] - \frac{2}{\alpha^3 \mu} \ln(1 + \alpha\mu) + \frac{1}{\alpha^2(1 + \alpha\mu)}. \quad (12)$$

where ψ' denotes the trigamma function.

For VELOVI, the Fisher information matrix was not derived analytically in this study. We therefore restricted the metric tensor analysis to SCVI and FlatVI. This analysis was primarily intended to assess whether the geometric regularization used by FlatVI results in a better-conditioned latent-space metric than SCVI.

For each latent representation, we computed the eigenvalues of $G(z)$ at sampled points in the latent space and quantified the local distortion using the condition number

$$\kappa(G(z)) = \frac{\lambda_{\max}(G(z))}{\lambda_{\min}(G(z))}. \quad (13)$$

The reported condition numbers correspond to the mean of $\kappa(G(z))$ over the sampled latent-space points.

Method	Latent dim.	$\kappa(G(z))$	α
FlatVI	10	11.86	1.55
FlatVI	20	20.76	0.82
FlatVI	50	4.18×10^{12}	0.28
FlatVI	100	3.50×10^{12}	0.11
FlatVI	200	3.71×10^{12}	0.05
scVI	10	524.20	–
scVI	20	3089.51	–
scVI	50	7.76×10^7	–
scVI	100	1.13×10^{20}	–
scVI	200	1.07×10^{20}	–

Table 10: Condition numbers of the metric tensor and learned α parameter for each model and latent dimension. Lower condition numbers indicate a better-conditioned local metric.

The results indicate that the condition number is substantially lower for FlatVI than for scVI for most latent dimensions, except at latent dimension 50. During training, FlatVI learns an α parameter that is used in the geometric regularization objective, $\|G(z) - \alpha I\|^2$. Interestingly, α decreases substantially as the latent dimension increases, approaching zero for the highest-dimensional representations.

This behavior may indicate that the regularization objective becomes increasingly difficult to balance with the reconstruction objective as the latent dimension increases. In this case, the learned value of α may reflect a trade-off between these two objectives. We did not investigate whether this behavior could be mitigated by modifying the regularization weight λ during training.

Despite this behavior, the condition number of the FlatVI metric remains substantially lower than that of scVI at the highest latent dimensions. These results suggest that the geometric regularization used by FlatVI can improve the conditioning of the local latent-space metric, although the effect of the regularization strength and its interaction with latent dimensionality would require further investigation.

Appendix D Clone Prediction Examples

For visualization purposes, we provide examples of clone-level predictions for each latent representation model using a latent dimension of 20. Predictions are shown at both t_4 and t_6 . We use UMAP to project the latent representations into two dimensions for visualization.

In each panel, gray points represent all real cells encoded into the corresponding latent space. The star marker indicates the initial state of the selected clone at t_2 , while blue points represent the observed cells from the same clone at the target time point (t_4 or t_6). Red points represent the cells simulated by scDiffEq from the initial clone state.

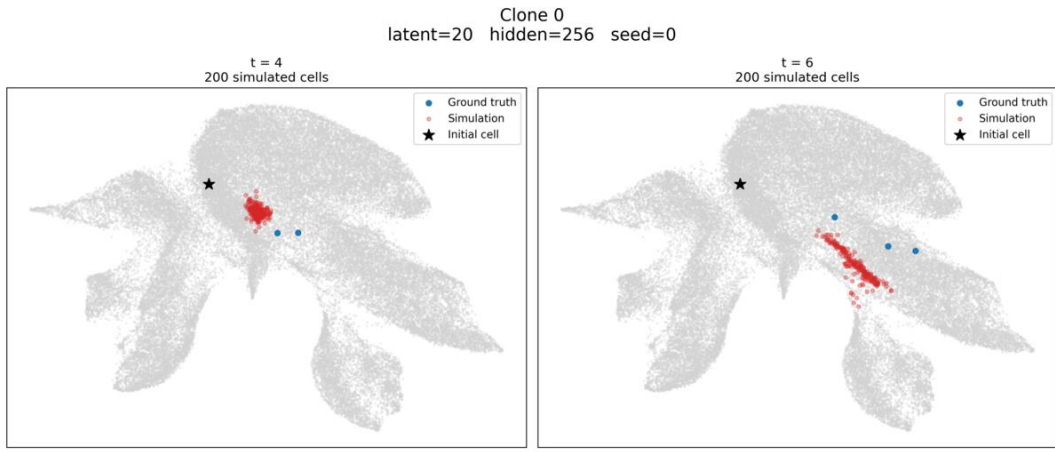


Figure 3: Example of a clone-level prediction using FlatVI with a latent dimension of 20.

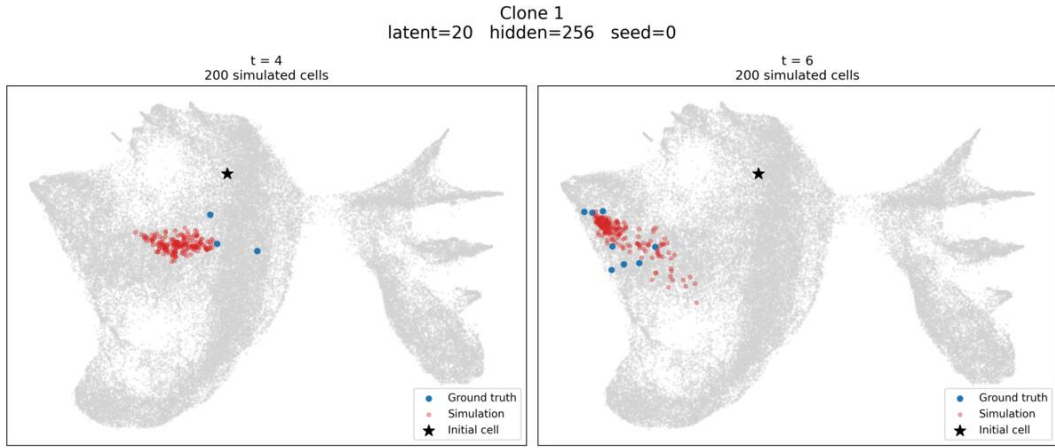


Figure 4: Example of a clone-level prediction using PCA with a latent dimension of 20.

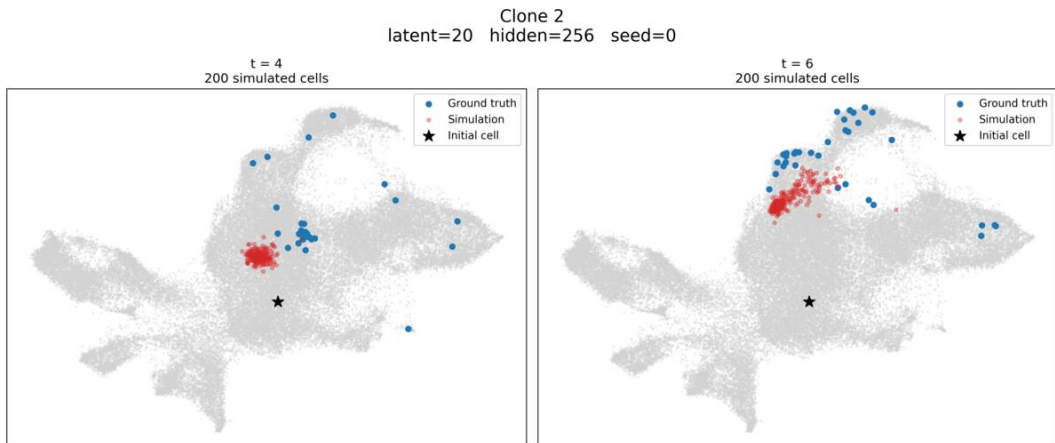


Figure 5: Example of a clone-level prediction using scVI with a latent dimension of 20.

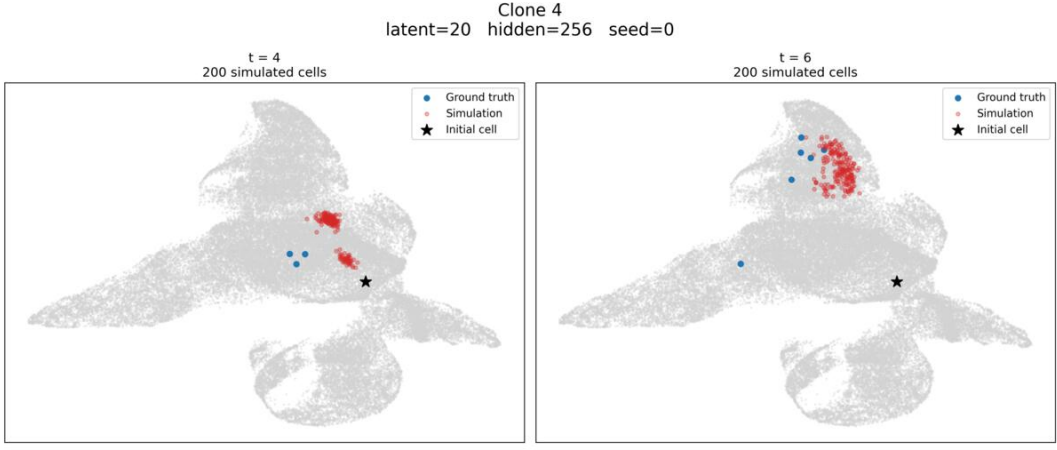


Figure 6: Example of a clone-level prediction using VELOVI with a latent dimension of 20.

Qualitatively, the simulated cells tend to be located closer to the observed final-state cells than to the initial clone state. However, when the observed final-state cells are broadly distributed in the latent space, the predictions appear more concentrated and tend to form a compact cluster near the center of the observed final-state population.

This behavior suggests that scDiffEq may have difficulty reproducing highly dispersed future-state populations at the clone level. Instead, the model may produce a more concentrated distribution that captures part of the central tendency of the observed future population. This observation raises the possibility that latent representations that preserve the local structure and dispersion of future clone states may facilitate more accurate clone-level predictions. However, this hypothesis would require a dedicated quantitative analysis to be confirmed.

Appendix E Study of the OT vs. drift-to-diffusion ratio

Since we do not impose a target drift-to-diffusion ratio during the training of our scDiffEq models, we investigate whether a lower ratio is associated with lower normalized OT in latent space.

For each scDiffEq model, we record the means and variances of the drift and diffusion values encountered during the final evaluation at t_4 and t_6 . We then aggregate these statistics for the best-performing model of each representation method and latent dimension.

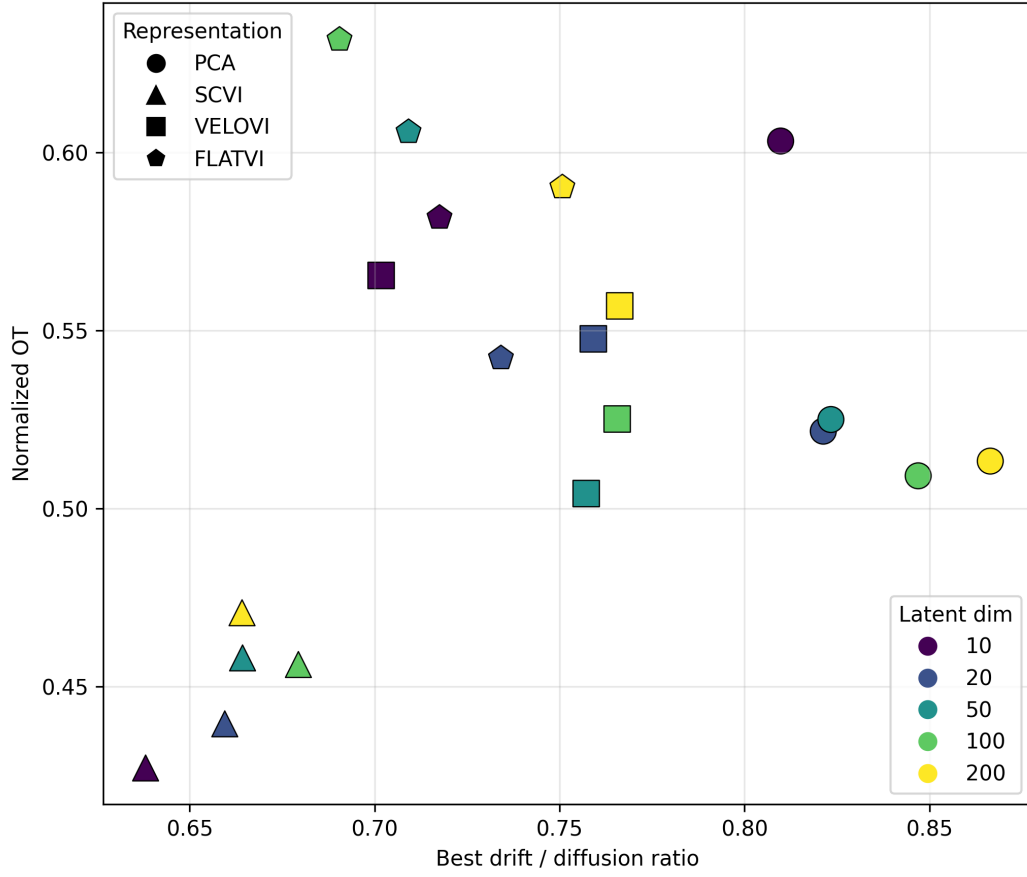


Figure 7: Normalized OT versus the drift-to-diffusion ratio at t_6 . Each shape represents a latent representation method, while each color corresponds to a latent dimensionality.

Qualitatively, the scVI representation stands out from the other representations, as all of its configurations are located in the lower-left region of the plot. This corresponds to both lower normalized OT and a lower drift-to-diffusion ratio.

For the other representation methods, no clear pattern of relative performance can be observed. Moreover, no clear correlation between normalized OT and the drift-to-diffusion ratio is apparent in the figure.

Both normalized OT and the drift-to-diffusion ratio appear to vary substantially across latent representations. However, this hypothesis would require a dedicated quantitative analysis to be confirmed.